

SpinoSarc: An open-source desktop tool for quantitative paraspinal muscle and dural sac analysis on lumbar spine MRI

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Software

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Summary

SpinoSarc is an open-source desktop application for quantitative analysis of paraspinal musculature and the dural sac on lumbar spine magnetic resonance imaging (MRI). It provides an end-to-end pipeline that ingests DICOM data directly from clinical scanners (Siemens, Philips, GE), performs automated segmentation of eight paraspinal muscle classes via the MuscleMap U-Net backbone ([MuscleMap contributors, 2023](#)), computes established quantitative metrics (cross-sectional area, fat fraction, total psoas area, psoas muscle index), supports manual region-of-interest measurement of dural sac cross-sectional area, and generates one-click PDF and Excel reports.

The application is implemented as a native macOS desktop tool with a PyQt6 graphical user interface and runs fully offline on Apple Silicon hardware. No imaging data leaves the local device, an important property for clinical research environments subject to patient privacy regulations. The interface is bilingual (English/Turkish).

SpinoSarc is intended for research use only and has not received regulatory approval. Quantitative metrics are reported alongside published literature reference values ([Barz et al., 2010](#); [Englesbe et al., 2010](#); [Hamaguchi et al., 2016](#)), but the software deliberately does not perform automated risk classification, leaving clinical interpretation to qualified experts.

Statement of need

Paraspinal muscle quality is an emerging biomarker in lumbar spine research, with growing evidence linking sarcopenia and muscle fat infiltration to chronic low back pain, surgical outcomes, and degenerative spinal pathology ([Hyun et al., 2016](#); [Shahidi et al., 2017](#)). Quantitative assessment, however, remains uncommon in clinical practice because existing tools require substantial technical infrastructure: dedicated workstations, command-line proficiency, manual NIFTI conversion of DICOM data, or institutional collaborations with research engineers.

MuscleMap ([MuscleMap contributors, 2023](#)) provides a high-quality open-source U-Net for paraspinal muscle segmentation but is implemented as a Python library that consumes NIFTI input and emits segmentation masks. Substantial additional engineering is required to use it in clinical research workflows: parsing multi-vendor DICOM series, aligning axial and sagittal coordinate systems for level identification, computing area and fat-fraction metrics, integrating manual dural sac measurements, and producing structured reports. These gaps create a barrier for clinicians who want to apply quantitative paraspinal analysis to research questions without becoming software engineers.

SpinoSarc closes this gap by wrapping MuscleMap in a complete clinical research workflow:

- 40 ■ **DICOM ingestion with vendor support.** The tool reads multi-slice axial and sagittal series
41 directly from clinical DICOM folders, including Philips JPEG Lossless compressed data,
42 and uses dcm2niix (Li et al., 2016) to produce consistent NIfTI input for segmentation.
43 Slices are sorted by InstanceNumber to match the behavior of clinical PACS viewers.
 - 44 ■ **Anatomical coordinate alignment.** Axial and sagittal series are cross-linked through
45 DICOM world coordinates (ImagePositionPatient), allowing the sagittal locator to
46 update with axial navigation and vice versa. This is essential for identifying the lumbar
47 level being analyzed.
 - 48 ■ **Quantitative metric extraction.** For each analyzed slice, SpinoSarc computes per-muscle
49 cross-sectional area and intensity-based fat fraction, total psoas area, and (when height
50 is provided) psoas muscle index. Reference values from established cohorts (Englesbe et
51 al., 2010; Hamaguchi et al., 2016) are displayed alongside results without automated
52 classification.
 - 53 ■ **Manual dural sac CSA.** A polygon region-of-interest tool allows the user to delineate
54 the dural sac on the axial slice. Cross-sectional area is computed in mm² with literature
55 thresholds for spinal stenosis (Barz et al., 2010) displayed as context.
 - 56 ■ **Reporting.** Results are exported as a single-page PDF report and a structured
57 Excel workbook containing per-muscle metrics, summary measurements, and patient
58 demographics.
 - 59 ■ **Native desktop distribution.** SpinoSarc is distributed as a signed-free .dmg installer for
60 Apple Silicon Macs. No Python environment, command-line tools, or internet connection
61 is required at runtime.
- 62 To our knowledge, SpinoSarc is the first open-source desktop application providing end-to-end
63 DICOM-to-report quantitative paraspinal analysis aimed at clinical researchers in radiology
64 and spine medicine.

65 Implementation

66 The application is written in Python 3.11 with PyQt6 for the graphical interface, MONAI
67 (Cardoso et al., 2022) and PyTorch (Paszke et al., 2019) for inference, pydicom (Mason &
68 others, n.d.) with pylibjpeg for DICOM parsing including compressed transfer syntaxes,
69 nibabel (Brett & others, n.d.) for NIfTI handling, and dcm2niix (Li et al., 2016) for DICOM-
70 to-NIfTI conversion. The macOS application is built with PyInstaller and distributed as a
71 self-contained .app bundle including the MuscleMap pretrained weights.

72 A typical analysis on Apple Silicon hardware (M-series, MPS backend) completes in 10–30
73 seconds per slice.

74 Limitations and future work

75 SpinoSarc currently supports macOS Apple Silicon only; Linux and Windows builds are planned.
76 Anatomical vertebral level (L1–S1) is not automatically identified—the user must select the
77 slice of interest, typically guided by the sagittal locator. Future versions will include automated
78 level detection and longitudinal tracking of repeated examinations.

79 The segmentation backbone (MuscleMap) was trained on a specific dataset and may not
80 generalize to all imaging protocols, scanners, or patient populations. Clinical validation
81 in independent cohorts is recommended before use of SpinoSarc outputs in any research
82 conclusion.

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